Section 3.5 – 3.6 Exercises

Directions: For exercises 1 - 11, divide the polynomials. Specify the quotient and the remainder.

1. $(x^2 + 5x - 1) \div (x - 1)$

2. $(2x^2 - 9x - 5) \div (x - 5)$

3. $(3x^2 + 23x + 14) \div (x + 7)$

4. $(4x^2 - 10x + 6) \div (4x + 2)$

5. $(6x^2 - 25x - 25) \div (6x + 5)$

6. $(-x^2 - 1) \div (x + 1)$

7. $(2x^2 - 3x + 2) \div (x + 2)$

8. $(x^3 - 126) \div (x - 5)$

9. $(3x^2 - 5x + 4) \div (3x + 1)$

10. $(x^3 - 3x^2 + 5x - 6) \div (x - 2)$

11. $(2x^3 + 3x^2 - 4x + 15) \div (x + 3)$

Directions: For exercises 12 - 18, use the Remainder Theorem to evaluate the function.

12. $(x^4 - 9x^2 + 14) \div (x - 2)$

13. $(3x^3 - 2x^2 + x - 4) \div (x + 3)$

14. $(x^4 + 5x^3 - 4x - 17) \div (x + 1)$

15. $(-3x^2 + 6x + 24) \div (x - 4)$

16. $(x^4 - 1) \div (x - 4)$

17. $(3x^3 + 4x^2 - 8x + 2) \div (x - 3)$

18. $(4x^3 + 5x^2 - 2x + 7) \div (x + 2)$

Directions: For exercises 19 – 26, use the given factor and the Factor Theorem to find all real zeros for the given polynomial function.

19. $f(x) = 2x^3 - 9x^2 + 13x - 6;\ x - 1$

20. $f(x) = 2x^3 + x^2 - 5x + 2;\ x + 2$

21. $f(x) = 3x^3 + x^2 - 20x + 12;\ x + 3$

22. $f(x) = 2x^3 + 3x^2 + x + 6;\ x + 2$

23. $f(x) = -5x^3 + 16x^2 - 9;\ x - 3$

24. $x^3 + 3x^2 + 4x + 12;\ x + 3$

25. $4x^3 - 7x + 3;\ x - 1$

26. $2x^3 + 5x^2 - 12x - 30, \ 2x + 5$

Directions: For exercises 27 - 43, use the Rational Zero Theorem to find all real zeros.

27. $x^3 - 3x^2 - 10x + 24 = 0$

28. $2x^3 + 7x^2 - 10x - 24 = 0$

29. $x^3 + 2x^2 - 9x - 18 = 0$

30. $x^3 + 5x^2 - 16x - 80 = 0$

31. $x^3 - 3x^2 - 25x + 75 = 0$

32. $2x^3 - 3x^2 - 32x - 15 = 0$

33. $2x^3 + x^2 - 7x - 6 = 0$

34. $2x^3 - 3x^2 - x + 1 = 0$

35. $3x^3 - x^2 - 11x - 6 = 0$

36. $2x^3 - 5x^2 + 9x - 9 = 0$

37. $2x^3 - 3x^2 + 4x + 3 = 0$

38. $x^4 - 2x^3 - 7x^2 + 8x + 12 = 0$

39. $x^4 + 2x^3 - 9x^2 - 2x + 8 = 0$

40. $4x^4 + 4x^3 - 25x^2 - x + 6 = 0$

41. $2x^4 - 3x^3 - 15x^2 + 32x - 12 = 0$

42. $x^4 + 2x^3 - 4x^2 - 10x - 5 = 0$

43. $4x^3 - 3x + 1 = 0$